

Analyzing Benchmarking Data from COCO's BBOB Suites

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- 1 Introduction
- 2 Background and Theoretical Framework
- 3 Performance Indicators
- 4 Benchmark Test Suites
- 5 Implementation
- 6 Results
 - BBOB Noiseless Results
 - BBOB Noisy Results
 - BBOB Bi-objective Results
- 7 Conclusion

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Introduction

- Problem: The COCO (COmparing Continuous Optimizers)archive contains hundreds of algorithms. How can we extract a small, reliable set of top-performing methods?

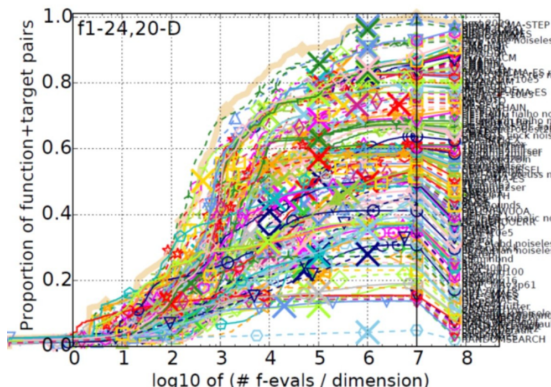


Figure: Algorithms from the years 2009-2015

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Black-Box Optimization

- Many real-world optimization problems rely on:
 - complex simulations
 - noisy physical or experimental measurements
 - computationally expensive evaluations
- Optimization problems where the internal structure of the objective function is unknown.
- No access to gradients, smoothness, or analytical expressions
- Algorithms rely solely on sequential **function evaluations**:

$$f(x_1), f(x_2), \dots, f(x_t)$$

The COCO Benchmarking Platform

- COCO provides standardized black-box optimization benchmarks
- Algorithms are evaluated on:
 - Multiple problem dimensions (smallest: $n = 2$ largest: $n = 40$)
 - Multiple function instances (shifted, rotated, rescaled).
 - **Goal:** Remove any lucky behavior such as accidental symmetries or coordinate alignments.
 - Predefined target precisions
- Performance measured using Expected Running Time (ERT)
- Public archive containing results for hundreds of algorithms across many years

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Target Precision and Success Criterion

- Performance is evaluated with respect to predefined target precisions Δf
- A target precision corresponds to reaching

$$f(x) - f^* \leq \Delta f$$

where f^* is the known optimal value

- For each function, multiple target precisions are defined, ranging from coarse to very fine accuracy
- A run is **successful** if it reaches the target precision before the evaluation budget is exhausted
- Runs that do not reach the target are treated as failures

Expected Running Time (ERT)

- Main performance indicator used in COCO benchmarks

Definition

For a given target precision Δf , the Expected Running Time (ERT) is defined as

$$\text{ERT}(\Delta f) = \frac{\sum_{i=1}^{15} RT_i}{\#\text{successes}},$$

where RT_i is the number of evaluations in run i .

- If no run reaches the target, $\text{ERT}(\Delta f) = +\infty$
- Lower ERT indicates faster convergence

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- Among several BBOB Test Suites, we will work with the ones with the largest data set of algorithms.
- The 3 different suites:
 - **BBOB Noiseless**
 - **BBOB Noisy**
 - **BBOB Bi-Objective**

- Baseline benchmark without stochastic perturbations
- Optimization difficulty arises solely from landscape properties
- Covers multiple dimensions and diverse function structures
- Serves as a reference case for algorithm performance

BBOB Noisy Test Suite

- Extends noiseless functions with stochastic perturbations
- Models uncertainty in real-world evaluations
- Noise types include light-tailed and heavy-tailed distributions
- Evaluates robustness and stability of algorithms

- A bi-objective optimization problem consists in simultaneously minimizing two objective functions:

$$f_1(x), f_2(x)$$

- **Pareto dominance:** a solution x dominates a solution y if

$$f_1(x) \leq f_1(y), \quad f_2(x) \leq f_2(y),$$

and at least one inequality is strict

- **Pareto front:** the set of all non-dominated solutions
- Represents the best achievable trade-offs between objectives

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Implementation Overview

- Input data: COCO benchmarking results
- Data indexed by:
 - year, dimension, function, target precision
- Performance metric: Expected Running Time (ERT)
- Goal: extract a small set of top-performing algorithms
- Two complementary selection strategies are implemented

Method 1: Absolute Best Algorithm

Definition

For a given dimension d , function f , and target precision Δf , the *absolute best algorithm* is the algorithm with the smallest Expected Running Time (ERT).

- Strict winner-based selection
- Aggregation across:
 - years
 - functions
 - target precisions
- Output: ranking of algorithms by number of wins per dimension

Method 2: Factor-Based Near-Optimality

- **Motivation:** strict winners may hide/overshadow similarly performing algorithms

Definition

Let $\text{ERT}_{\min}(d, f, \Delta f)$ be the minimal ERT for a given dimension d , function f , and target precision Δf . An algorithm A is *near-optimal* if

$$\text{ERT}_A(d, f, \Delta f) \leq \alpha \cdot \text{ERT}_{\min}(d, f, \Delta f),$$

for a user-chosen factor $\alpha > 1$.

- α controls selection strictness
- All near-optimal algorithms are counted
- **Output:** frequency-based ranking of consistently strong algorithms

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BBOB Noiseless Results

- For each $(d, f, \Delta f)$, the algorithm with the smallest ERT is selected
- Strict winner-based selection
- Counts are aggregated across functions, targets, and years

dimension	2	3	5	10	20	40
rank						
1	(HMLSL_pal, 16)	(lq-CMA-ES_Hansen, 12)	(HE-ES_Glasmachers, 15)	(SLSQP+Iq-CMA-ES_Hansen, 12)	(CMAES-APOP-KMA_Nguyen, 12)	(SLSQP-scipy-2019_Varelas, 14)
2	(DE-BFGS_vogliss, 9)	(HMLSL_pal, 9)	(OQNLP_pal, 12)	(BIPOP-saACM-k_loschilov, 9)	(BIPOP-saACM-k_loschilov, 11)	(BIPOP-aCMA-STEP_loschilov, 11)
3	(NELDERDOERR_doerr, 8)	(fminunc_pal, 9)	(BIPOP-aCMA-STEP_loschilov, 7)	(BIRMIN_Kudela, 8)	(NIPOPaCMA_loschilov, 10)	(SLSQP+Iq-CMA-ES_Hansen, 11)
4	(DTS-CMA-ES_Pitra, 7)	(OQNLP_pal, 7)	(HMLSL_pal, 7)	(BIPOP-aCMA-STEP_loschilov, 6)	(L-BFGS-B-scipy-2019_Varelas, 9)	(NEUWOA_ros, 10)
5	(SHADE-LM-POP4-to-10_Okulewicz, 6)	(NELDERDOERR_doerr, 6)	(PSA-CMA-ES_Nishida, 7)	(FULLNEUWOA_ros, 6)	(SLSQP-scipy-2019_Varelas, 9)	(BIPOP-saACM-k_loschilov, 9)

Figure: BBOB Suite: Ranking of Algorithms per Dimension for Method 1, Showing (Algorithm, Number of Wins).

BBOB Noiseless Method 2

- Algorithms within a factor 2 of the best ERT are selected
- Aggregated across functions, targets, and years

rank	dim 2	dim 3	dim 5	dim 10	dim 20	dim 40
1	SLSQP+lq-CMA-ES_Hansen (52)	lq-CMA-ES_Hansen (54)	lq-CMA-ES_Hansen (59)	SLSQP-11-scipy_Hansen (39)	SLSQP+lq-CMA-ES_Hansen (41)	SLSQP+lq-CMA-ES_Hansen (43)
2	DIRECT-REV_Kudela (51)	SLSQP+lq-CMA-ES_Hansen (41)	SLSQP+lq-CMA-ES_Hansen (47)	SLSQP+lq-CMA-ES_Hansen (38)	IPOPsaACM_loschilov (36)	SLSQP-11-scipy_Hansen (34)
3	NELDERDOERR_doerr (45)	SLSQP-11-scipy_Hansen (37)	SLSQP-11-scipy_Hansen (46)	BIPOPsaACM_loschilov (35)	lq-CMA-ES_Hansen (30)	NEWUOA_ros (30)
4	SLSQP-11-scipy_Hansen (45)	DIRECT-REV_Kudela (36)	SLSQP-scipy-2019_Varelas (36)	lq-CMA-ES_Hansen (33)	SLSQP-11-scipy_Hansen (29)	adm-CMA-ES_Gissler (30)
5	lq-CMA-ES_Hansen (36)	SHADE-LM-POP4-to-10_Okulewicz (34)	CMA-ES-2019_Hansen (27)	IPOPsaACM_loschilov (27)	NIPOPaCMA_loschilov (27)	SLSQP-scipy-2019_Varelas (28)
6	DTS-CMA-ES_Pitra (34)	SLSQP-scipy-2019_Varelas (33)	SHADE-LM_Okulewicz (26)	adm-CMA-ES_Gissler (24)	PSA-CMA-ESwRS_Nishida (27)	NIPOPaCMA_loschilov (27)

Figure: Ranking of Algorithms per Dimension for the Factor Method, (Algorithm, Number of Wins)

- **Method 1 (Absolute Best):**

- Identifies strict winners for each $(d, f, \Delta f)$
- Highly sensitive to small ERT differences
- No algorithm consistently dominates across dimensions

- **Method 2 (Factor-Based):**

- Identifies algorithms that are near-optimal within a factor
- Includes more algorithms, but dominance remains unclear
- Confirms a highly competitive and fragmented landscape

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Method 1: BBOB Noisy Results

dimension	2	3	5	10	20	40
rank						
1	(IPOP-ACTCMA-ES_ros_noisy, 20)	(IPOPsaACM_loschilov_noisy, 29)	(IPOPsaACM_loschilov_noisy, 42)	(IPOPsaACM_loschilov_noisy, 36)	(IPOPsaACM_loschilov_noisy, 55)	(IPOP-ACTCMA-ES_ros_noisy, 53)
2	(FULLNEWUOA_ros_noisy, 12)	(IPOP-ACTCMA-ES_ros_noisy, 22)	(1komma4mirser_brockhoff_noisy, 18)	(1komma4mirser_brockhoff_noisy, 16)	(IPOP-CMA-ES_ros_noisy, 20)	(1komma4mirser_brockhoff_noisy, 37)
3	(IPOPsaACM_loschilov_noisy, 12)	(1komma4mirser_brockhoff_noisy, 17)	(IPOP-ACTCMA-ES_ros_noisy, 14)	(IPOP-ACTCMA-ES_ros_noisy, 13)	(1komma4mirser_brockhoff_noisy, 16)	(IPOP-CMA-ES_ros_noisy, 25)
4	(1komma4mirser_brockhoff_noisy, 11)	(PSAaLmD-CMA-ES_Nishida_bbob-noisy, 9)	(IPOP-CMA-ES_ros_noisy, 12)	(IPOP-CMA-ES_ros_noisy, 13)	(PSAaSmC-CMA-ES_Nishida_bbob-noisy, 14)	(CMAEGS_finck_noisy, 5)
5	(AMALGAM_bosman_noisy, 11)	(SNOBFIT_huyer_noisy, 9)	(PSAaLmD-CMA-ES_Nishida_bbob-noisy, 10)	(BIPOP-CMA-ES_hansen_noisy, 9)	(IPOP-ACTCMA-ES_ros_noisy, 11)	(SNES_schaul_noisy, 4)
6	(IPOP-CMA-ES_ros_noisy, 10)	(PSAaLmC-CMA-ES_Nishida_bbob-noisy, 8)	(SNES_schaul_noisy, 9)	(SNES_schaul_noisy, 9)	(RCGA_tran_noisy, 5)	(BIPOP-CMA-ES_hansen_noisy, 2)

Figure: Ranking of Algorithms per Dimension for the Method 1, showing (Algorithm, Number of Wins)

Method 2: BBOB Noisy Results

rank	dim 2	dim 3	dim 5	dim 10	dim 20	dim 40
1	IPOPsaACM_loschilov_noisy (78)	IPOPsaACM_loschilov_noisy (101)	IPOPsaACM_loschilov_noisy (97)	IPOPsaACM_loschilov_noisy (101)	IPOPsaACM_loschilov_noisy (115)	IPOP-ACTCMA-ES_ros_noisy (102)
2	IPOP-ACTCMA-ES_ros_noisy (68)	IPOP-ACTCMA-ES_ros_noisy (88)	IPOP-ACTCMA-ES_ros_noisy (80)	IPOP-ACTCMA-ES_ros_noisy (87)	IPOP-ACTCMA-ES_ros_noisy (97)	IPOP-CMA-ES_ros_noisy (69)
3	IPOP-CMA-ES_ros_noisy (53)	IPOP-CMA-ES_ros_noisy (50)	IPOP-CMA-ES_ros_noisy (53)	IPOP-CMA-ES_ros_noisy (51)	IPOP-CMA-ES_ros_noisy (51)	BIPOP-CMA-ES_hansen_noisy (64)
4	BIPOP-CMA-ES_hansen_noisy (45)	BIPOP-CMA-ES_hansen_noisy (47)	BIPOP-CMA-ES_hansen_noisy (44)	BIPOP-CMA-ES_hansen_noisy (44)	BIPOP-CMA-ES_hansen_noisy (42)	1komma4mirser_brockhoff_noisy (41)
5	SNES_schau1_noisy (40)	PSAaLmC-CMA-ES_Nishida_bbob-noisy (41)	PSAaLmC-CMA-ES_Nishida_bbob-noisy (37)	PSAaSmD-CMA-ES_Nishida_bbob-noisy (39)	1komma4mirser_brockhoff_noisy (42)	IPOP-SEP-CMA-ES_ros_noisy (35)
6	1komma4mirser_brockhoff_noisy (39)	1komma4mirser_brockhoff_noisy (38)	PSAaLmD-CMA-ES_Nishida_bbob-noisy (31)	PSAaSmC-CMA-ES_Nishida_bbob-noisy (37)	PSAaSmC-CMA-ES_Nishida_bbob-noisy (32)	SNES_schau1_noisy (10)

Figure: BBOB Noisy: Ranking of Algorithms per Dimension for the Factor Method, Showing (Algorithm, Number of Wins)

- **Method 1 (Absolute Best):**

- Identifies strict winners for each $(d, f, \Delta f)$
- Clear dominance patterns emerge across dimensions
- A small set of algorithms consistently outperforms others

- **Method 2 (Factor-Based):**

- Identifies near-optimal algorithms within a factor
- Counts are inflated: they used to share the count, now they both get counted as best algorithms
- Reinforces the same dominant algorithms
- Confirms robustness of these algorithms across all dimensions

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Method 1: BBOB Bi-objective Results

dimension	2	3	5	10	20	40
rank						
1	(SPEA2-platypus_Brockhoff_bbob-biobj, 53)	(HMO-CMA-ES_Loshchilov_bbob-biobj, 82)	(HMO-CMA-ES_Loshchilov_bbob-biobj, 99)	(HMO-CMA-ES_Loshchilov_bbob-biobj, 125)	(HMO-CMA-ES_Loshchilov_bbob-biobj, 131)	(GDE3-platypus_Brockhoff_bbob-biobj, 109)
2	(UP-MO-CMA-ES_Krause_bbob-biobj, 51)	(SPEA2-platypus_Brockhoff_bbob-biobj, 38)	(UP-MO-CMA-ES_Krause_bbob-biobj, 38)	(UP-MO-CMA-ES_Krause_bbob-biobj, 20)	(TPB_Tanabe, 21)	(RM-MEDA_Auger_bbob-biobj, 77)
3	(HMO-CMA-ES_Loshchilov_bbob-biobj, 50)	(UP-MO-CMA-ES_Krause_bbob-biobj, 33)	(SPEA2-platypus_Brockhoff_bbob-biobj, 30)	(TPB_Tanabe, 19)	(UP-MO-CMA-ES_Krause_bbob-biobj, 21)	(SPEA2-platypus_Brockhoff_bbob-biobj, 26)
4	(K-RVEA_Tanabe, 26)	(K-RVEA_Tanabe, 20)	(TPB_Tanabe, 19)	(IBEA-platypus_Brockhoff_bbob-biobj, 18)	(GDE3-platypus_Brockhoff_bbob-biobj, 19)	(MOEAD-platypus_Brockhoff_bbob-biobj, 9)
5	(IBEA-platypus_Brockhoff_bbob-biobj, 16)	(TPB_Tanabe, 18)	(GDE3-platypus_Brockhoff_bbob-biobj, 17)	(SPEA2-platypus_Brockhoff_bbob-biobj, 18)	(SPEA2-platypus_Brockhoff_bbob-biobj, 14)	(SMS-EMOA-DE_Auger_bbob-biobj, 1)

Figure: BBOB Bi-Objective: Table showing the Absolute Best Top 5 algorithms per Dimension

Method 2: BBOB Bi-Objective Results

rank	dim 2	dim 3	dim 5	dim 10	dim 20	dim 40
1	HMO-CMA-ES_Loshchilov_bbob-biobj (125)	HMO-CMA-ES_Loshchilov_bbob-biobj (135)	HMO-CMA-ES_Loshchilov_bbob-biobj (141)	HMO-CMA-ES_Loshchilov_bbob-biobj (152)	HMO-CMA-ES_Loshchilov_bbob-biobj (159)	GDE3-platypus_Brockhoff_bbob-biobj (130)
2	MO-DIRECT-HV-Rank_Wong_bbob-biobj (74)	SPEA2-platypus_Brockhoff_bbob-biobj (55)	UP-MO-CMA-ES_Krause_bbob-biobj (48)	TPB_Tanabe (45)	TPB_Tanabe (42)	RM-MEDA_Auger_bbob-biobj (96)
3	UP-MO-CMA-ES_Krause_bbob-biobj (68)	K-RVEA_Tanabe (53)	TPB_Tanabe (41)	UP-MO-CMA-ES_Krause_bbob-biobj (38)	UP-MO-CMA-ES_Krause_bbob-biobj (34)	SPEA2-platypus_Brockhoff_bbob-biobj (62)
4	DEMO_Tusar_bbob-biobj (63)	UP-MO-CMA-ES_Krause_bbob-biobj (49)	SPEA2-platypus_Brockhoff_bbob-biobj (39)	GDE3-platypus_Brockhoff_bbob-biobj (38)	GDE3-platypus_Brockhoff_bbob-biobj (31)	MOEAD-platypus_Brockhoff_bbob-biobj (24)
5	SPEA2-platypus_Brockhoff_bbob-biobj (61)	TPB_Tanabe (46)	K-RVEA_Tanabe (37)	COMO-316_dufosse_bbob-biobj (32)	SPEA2-platypus_Brockhoff_bbob-biobj (23)	SMS-EMOADE_Auger_bbob-biobj (1)

Figure: BBOB Bi-Objective: Ranking of Algorithms per Dimension for the Factor method, Showing (Algorithm, Number of Wins)

- **Method 1 (Absolute Best):**

- Identifies strict winners based on Pareto-based performance
- Strong dominance patterns persist across all dimensions
- The same algorithms remain top-ranked for all dimensions

- **Method 2 (Factor-Based):**

- Identifies near-optimal algorithms within a factor 2.
- Yields nearly identical rankings to Method 1
- Confirms a clear and robust hierarchy of algorithms

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Main Contributions and Findings

- Developed a **filtering framework** to exploit large COCO benchmarking archives
- Two complementary selection strategies:
 - strict winner-based selection
 - factor-based near-optimality selection
- Revealed distinct algorithm-selection:
 - **Noiseless:** highly competitive landscape, no clear winner
 - **Noisy:** dominance of a small set of robust algorithms
 - **Bi-objective:** strong and stable hierarchy across dimensions

Practical Implications

- Reduces over 250 algorithms to a short, reliable list
- Enables informed algorithm selection
- Saves benchmarking time and resources

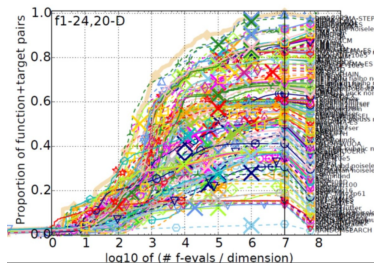


Figure: Years 2009-2015 Before the Filtering approach to Reduce the Number of Algorithms for the BBOB Noiseless Test Suite

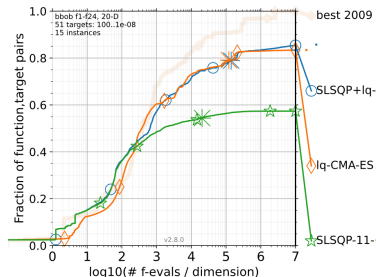


Figure: Top 3 Algorithms for the Noiseless 20D Test Suite after Filtering